Physics 4700H: Assignment V

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Problem 1 (Kennett problem 2.6)**.** Consider N distinguishable atoms with two energy levels $\pm\epsilon$. Assume that there are very weak interactions between the atoms such that equilibrium can be reached but that these can be ignored when calculating equilibrium properties. Let E be the total energy of the system.

(a) What is the entropy of the system as a function of E?

(b) What is the temperature of the system as a function of energy?

(c) What are the temperatures corresponding to $U = 0$ and to $U = -N\epsilon$? Comment on your results.

Solution 1.

(a) We write $E = N_+\epsilon + N_-(-\epsilon) = \epsilon\,(N_+ - N_-)$ where N_+ and N_- represent the number of particles in the ϵ and $-\epsilon$ states respectively, and hence $N_+ + N_- = N$. Such a system has fixed energy (along with volume and number) and hence is in the microcanonical ensemble where we may write entropy as $S_B = k_B \ln(\Omega)$ and since there are only two available states,

$$\Omega = \frac{N!}{\left[\frac{1}{2}\left(N + \frac{E}{\epsilon}\right)\right]!\left[\frac{1}{2}\left(N - \frac{E}{\epsilon}\right)\right]!}$$

where we've taken the usual $X = N_+ - N_-$ and used energy. This gives an entropy of

$$S_B = k_B \ln\left(\frac{N!}{\left[\frac{1}{2}\left(N + \frac{E}{\epsilon}\right)\right]!\left[\frac{1}{2}\left(N - \frac{E}{\epsilon}\right)\right]!}\right).$$

(b) To do this we consider the thermodynamic identity at constant volume, $dE = TdS$ which gives us an equation for temperature,

$$\frac{1}{T} = \frac{\partial S}{\partial E}\bigg)_{N,V}.$$

To differentiate the expression we found in (a), unless we want apply the Gamma Function, we must simplify the expression to eliminate the factorials. To do this, we break up the ln, then apply Sterling's approximation, $\ln(N!) \approx N\ln(N) - N$,

$$\begin{aligned}
\frac{S}{k_B} &= \ln\left(\frac{N!}{\left[\frac{1}{2}\left(N + \frac{E}{\epsilon}\right)\right]!\left[\frac{1}{2}\left(N - \frac{E}{\epsilon}\right)\right]!}\right) \\
&= \ln(N!) - \ln\left(\left[\frac{1}{2}\left(N + \frac{E}{\epsilon}\right)\right]!\right) - \ln\left(\left[\frac{1}{2}\left(N - \frac{E}{\epsilon}\right)\right]!\right) \\
&\approx N\ln(N) - \cancel{N} \\
&\quad - \left[\frac{1}{2}\left(N + \frac{E}{\epsilon}\right)\right]\ln\left(\frac{1}{2}\left(N + \frac{E}{\epsilon}\right)\right) + \frac{1}{2}\left(\cancel{N} + \cancel{\frac{E}{\epsilon}}\right) \\
&\quad - \left[\frac{1}{2}\left(N - \frac{E}{\epsilon}\right)\right]\ln\left(\frac{1}{2}\left(N - \frac{E}{\epsilon}\right)\right) + \frac{1}{2}\left(\cancel{N} - \cancel{\frac{E}{\epsilon}}\right) \\
&= N\ln(N) \\
&\quad - \left[\frac{1}{2}\left(N + \frac{E}{\epsilon}\right)\right]\ln\left(\frac{1}{2}\left(N + \frac{E}{\epsilon}\right)\right) \\
&\quad - \left[\frac{1}{2}\left(N - \frac{E}{\epsilon}\right)\right]\ln\left(\frac{1}{2}\left(N - \frac{E}{\epsilon}\right)\right).
\end{aligned}$$

Now we consider the derivative with respect to E,

$$\begin{aligned}\frac{1}{k_B}\frac{\partial S}{\partial E} = \frac{1}{k_B T} &\approx -\frac{\partial}{\partial E}\left[\frac{1}{2}\left(N+\frac{E}{\epsilon}\right)\right]\ln\left(\frac{1}{2}\left(N+\frac{E}{\epsilon}\right)\right) - \frac{\partial}{\partial E}\left[\frac{1}{2}\left(N-\frac{E}{\epsilon}\right)\right]\ln\left(\frac{1}{2}\left(N-\frac{E}{\epsilon}\right)\right)\\ &= -\frac{1}{2\epsilon}\ln\left(\frac{1}{2}\left(N+\frac{E}{\epsilon}\right)\right) - \cancel{\left[\frac{1}{2}\left(N+\frac{E}{\epsilon}\right)\right]\left(\frac{1}{2\epsilon}\right)\frac{2}{N+\frac{E}{\epsilon}}}\\ &\quad + \frac{1}{2\epsilon}\ln\left(\frac{1}{2}\left(N-\frac{E}{\epsilon}\right)\right) + \cancel{\left[\frac{1}{2}\left(N-\frac{E}{\epsilon}\right)\right]\left(\frac{1}{2\epsilon}\right)\frac{2}{N-\frac{E}{\epsilon}}}\\ &= \frac{1}{2\epsilon}\left[\ln\left(\frac{N-\frac{E}{\epsilon}}{N+\frac{E}{\epsilon}}\right)\right]\end{aligned}$$

and so

$$T(E) = \frac{2\epsilon k_B}{\ln\left(\frac{N-\frac{E}{\epsilon}}{N+\frac{E}{\epsilon}}\right)}.$$

(c) At $E = 0$,

$$T(E=0) = \frac{2\epsilon k_B}{\ln\left(\frac{N}{N}\right)} \to \infty$$

as $\ln(N/N) = \ln(1) = 0$. At $E = -N\epsilon$,

$$T(E=-N\epsilon) = \frac{2\epsilon k_B}{\ln\left(\frac{-2N}{N-N}\right)} \to 0$$

as $\ln(\infty) \to \infty$. These, somewhat unintuitively due to my desire to make an analogy to a kinetic system with a bunch of particles whizzing around, make sense. At $E = -N\epsilon$, the system is in its lowest energy configuration, analogous to all the particles not moving, fine. At $E = 0$, the system is actually *not* in its lowest energy state, in fact it is in a state where nothing is known about the states of the individual particles, a state of maximum entropy.

Problem 2. Show that the internal energy may be written in terms of the partition function as follows,

$$U = k_B T^2 \frac{\partial \ln Z}{\partial T}.$$

Solution 2. We know that

$$Z = \sum_i e^{\epsilon_i / k_B T}.$$

This lets us write that

$$\frac{\partial \ln Z}{\partial T} = \frac{\partial Z}{\partial T}\frac{\partial \ln(Z)}{\partial Z} = -\frac{1}{k_B T^2}\sum_i \epsilon_i e^{\epsilon_i/k_B T}\frac{1}{Z}.$$

Recalling that $p_i = e^{\epsilon_i/k_B T}/Z$, the expression can be written as

$$\frac{\partial \ln Z}{\partial T} = \frac{1}{k_B T^2}\sum_i \epsilon_i p_i = \frac{U}{k_B T^2}.$$

Multiplying this by $k_B T^2$ we obtain the result,

$$U = k_B T^2 \frac{\partial \ln Z}{\partial T}.$$